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Digital Material Passport

ID	DMP-METAL-005	Version	1.0.0
Issue Date	2025-05-17	Certificate Type	EN 10204 3.1

Business Transaction

Order		Delivery	
Order ID	PO-23456	Delivery ID	DN-65432
Position	1	Position	1
Date	2025-05-01	Date	2025-05-16
Quantity	1000 kg	Quantity	1000 kg

Product Information

Product Name	Titanium Alloy Ti-6Al-4V ELI
Batch ID	T-65432-01
Surface Condition	Machined
Production Date	2025-05-15
Country of Origin	DE

Product Norms

Standard	ASTM F136 (2022)
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Material Designations

Number (UNS)	R56401
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Product Shape

Form	RoundBar
Length	3000 mm
Diameter	30 mm

Chemical Analysis

Heat Number	T-65432
Melting Process	VAR
Casting Date	2025-05-14
Casting Method	VacuumCasting
Sample Location	Product

Elements

Symbol	Ti	Al	V	Fe	O	C	N	H
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<i>Unit</i>	%	%	%	%	%	%	%	ppm
<i>Min</i>	-	5.5	3.5	-	-	-	-	-
<i>Max</i>	-	6.5	4.5	0.25	0.13	0.08	0.05	120
<i>Actual</i>	89.32	6.02 ± 0.05	3.95 ± 0.03	0.18	0.11	0.026	0.012	35

Mechanical Properties

<i>Property</i>	<i>Symbol</i>	<i>Actual</i>	<i>Minimum</i>	<i>Maximum</i>	<i>Method</i>	<i>Status</i>
Fatigue Test Room temperature, R = 0.1					ASTM E466	✓
Cycles (N)	10000	50000	100000	500000	1000000	10000000
Value	650	635	610	590	570	550
Notch Sensitivity		0.85 - 0.92			Internal Method TS-5432	✓

Physical Properties

<i>Property</i>	<i>Symbol</i>	<i>Actual</i>	<i>Target/Min</i>	<i>Maximum</i>	<i>Method</i>	<i>Status</i>
Density		4.43 ± 0.01g/cm³	4.43g/cm³	-	ASTM B311	✓

Supplementary Tests

Property		Actual			Target/Min	Maximum	Method		Status
Microstructure Examination		Equiaxed alpha with intergranular beta			-	-	ASTM E407		✓
Ultrasonic Inspection		Yes No indications greater than reference standard			-	-	ASTM E2375		✓
Surface Quality Assessment		Class 1 - Medical Grade			-	-	Visual Inspection per ASTM F136		✓
Alpha Case Depth		5µm			-	25µm	Microhardness Traverse		✓
Grain Size Distribution		8 - 10ASTM No.			7 - 12ASTM No.	-	ASTM E112		✓
Hardness Profile		Array data (see below)			-	-	ASTM E384		✓
Distance from - surface (mm)	0.1	0.5	1	2	3	5	10	15	
Value [HV]	345	350	352	350	349	348	351	347	

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with ASTM F136 and meets all requirements for surgical implant applications.

Validated By

<i>Name</i>	<i>Title</i>	<i>Department</i>	<i>Date</i>
Dr. Markus Weber	Head of Metallurgy	Research & Quality	2025-05-17